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Relapse prevention study of paliperidone extended-release tablets in Chinese patients with schizophrenia☆,☆☆,★

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ABSTRACT

Objectives: The objective of this study was to evaluate the long-term efficacy, safety, and tolerability of paliperidone extended-release (pali ER), in Chinese patients with schizophrenia.

Methods: In this parallel-group, relapse prevention, phase-3 study (screening [14-day], pali ER open-label run-in [8-week] and stabilization [6-week] phases, and double-blind (DB) treatment [variable duration], and open-label extension phases [24-week]), 136/201 patients with schizophrenia were randomized (1:1) to pali ER (3–12 mg) or placebo during the DB phase.

Results: Final analysis showed that, out of 135 patients in ITT (DB) population, 71 (52.6%) had a relapse event, 45 (33.3%) were ongoing at the time the study was stopped, and 19 (14.1%) discontinued from the DB phase. Time to relapse (primary endpoint) favored pali ER (hazard ratio = 5.23 [95% CI: 2.96, 9.25], $p < 0.0001$). Rate of relapses (55/71 [77.5%] placebo; 16/64 [25%] pali ER) and secondary endpoints (change from baseline in Positive And Negative Syndrome Scale [PANSS] and Clinical Global Impression — Severity Scores) were significantly lower ($p < 0.001$) in pali ER group vs placebo, in favor of pali ER. More psychiatric-related treatment-emergent adverse events (TEAEs) occurred in placebo- (21.1%) than pali ER group (10.9%). Most common (>3%) TEAEs in placebo group were insomnia and schizophrenia (8.5% each), while in pali ER group were aggression and akathisia (4.7% each), and schizophrenia, tremor, nausea, amenorrhea, and salivary hypersecretion (3.1% each). All serious TEAEs were psychiatric-related (schizophrenia, aggression, completed suicide, auditory hallucination, suicide attempt) and more frequent in placebo- (11.3%) versus pali ER group (3.1%). Death and tardive dyskinesia-related discontinuation (n = 1 each) occurred in placebo group. Body weight increase from run-in baseline was greater in pali ER group (mean increase: 3.90 kg) versus placebo (mean increase: 2.05 kg).

Conclusions: This study confirms the findings from earlier pali ER global relapse-prevention studies and demonstrates that pali ER treatment (3–12 mg) is efficacious over the long-term and significantly delays relapse in Chinese patients with schizophrenia. No new safety signals were detected in this population.

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1. Introduction

Schizophrenia accounts for 55% of all psychotic disorders in the adult Chinese population (Li, 2011). Second generation antipsychotics (SGAs) are preferred as the primary treatment option in patients with schizophrenia, as they are effective in the treatment of both positive and negative symptoms (Csernansky et al., 2002; Kane et al., 2002; Lehman et al., 2004; Schooler et al., 2005; Taylor, 2003) and associated with fewer motor adverse effects. In China, the use of SGAs for the treatment of schizophrenia has increased from 53% in 1999 to 77% in 2008 and oral antipsychotics like olanzapine, quetiapine and risperidone are widely used (An et al., 2010).

Schizophrenia is a chronic condition requiring consistent, long-term treatment; it is common for patients to discontinue medication on their own (Kramer et al., 2007). Non-adherence to oral medications results in

Abbreviations: AE, Adverse event; AIMS, Abnormal involuntary movement scale; ANCOVA, Analysis of covariance; BARS, Barnes akathisia rating scale; BL, Baseline; BMI, Body mass index; CGI-S, Clinical global impression severity; CI, Confidence interval; C-SSRS, Columbia suicide severity rating scale; DB, Double blind; DSM-IV-TR, Diagnostic and statistical manual of mental disorders, 4th version; ECG, Electrocardiogram; EPS, Extrapyramidal symptom; ER, Extended release; HR, Hazards ratio; IDMC, Independent Data Monitoring Committee; ITT, Intent-to-treat; LOCF, Last observation carried forward; pali ER, Paliperidone extended-release; PANSS, Positive and negative syndrome scale; PSP, Personal and social performance scale; RI, Run in; SAS, Simpson–angus rating scale; SD, Standard deviation; SE, Standard error; SGAs, Second generation antipsychotics; ST, Stabilization; TEAEs, Treatment-emergent adverse events.

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increased relapse rates, which can subsequently lead to disease progression or even treatment failure (Emsley et al., 2008b). Thus, adherence to long-term treatment of schizophrenia remains a major treatment concern.

Paliperidone extended-release (pali ER), a SGA designed to deliver paliperidone at a relatively constant rate over a 24-hour period, is approved in the United States, European Union, and many other countries for the treatment of schizophrenia in adults (Invega product information, 2007). Pali ER is also approved in the United States and European Union as mono- and adjunctive therapy for the treatment of schizoaffective disorder (Invega product information, 2007). In completed clinical studies, pali ER 3 to 15 mg/day is efficacious and generally well-tolerated (Emsley et al., 2008a; Kane et al., 2007; Kramer et al., 2010). Asians generally accounted for a very small (≤5%) percentage of these study populations. It is well established that race and ethnic differences can influence treatment response, as well as the type and the extent of adverse events associated with antipsychotic treatment (Banerjee, 2012; Bhugra and Bhui, 1999; Coppola et al., 2012; Versola-Russo, 2006; Williams and Earl, 2007). Consequently, studies conducted in populations composed primarily of specific ethnic backgrounds or races are needed. The current study was hence conducted to confirm the efficacy of pali ER in delaying time to relapse and its overall safety in Chinese patients with schizophrenia.

2. Methods

2.1. Patients

The study was conducted at 18 sites within the People's Republic of China (from June 2011 to April 2013). Patients of either sex, aged ≥18 years, diagnosed with schizophrenia based on Diagnostic and Statistical Manual of Mental Disorders, 4th version (DSM-IV-TR) for at least 1 year before screening, and a Positive and Negative Syndrome Scale (PANSS) total score between 70 and 120 (inclusive), at screening and baseline were eligible for enrollment.

Major exclusion criteria for the study included: drug dependence (excluding nicotine and caffeine dependence) within 6 months before screening according to DSM-IV, history of cardiovascular, respiratory, neurologic, renal, hepatic, endocrine, or immunologic diseases, presence of circumstances that may increase the risk of the occurrence of torsade de pointes or sudden death, heart rate < 50 bpm, presence of congenital prolongation of the QT interval or demonstration of repeated prolonged QTc Fridericia interval > 450 ms in >1 electrocardiogram (ECG), neuroleptic malignant syndrome and hypersensitivity to risperidone, paliperidone, or their excipients. Patients treated with clozapine for treatment refractory or treatment resistant schizophrenia, monoamine oxidase inhibitor antidepressants within 4 weeks before screening, depot antipsychotic drugs within 120 days, paliperidone palmitate within 10 months or electroconvulsive therapy within 60 days before screening, and pregnant and lactating women were all excluded from the study.

The Independent Ethics Committee or Institutional Review Board at each study site approved the protocol. The study was conducted in accordance with the ethical principles that have their origin in the Declaration of Helsinki, consistent Good Clinical Practices, and applicable regulatory requirements. All participants provided written informed consent.

2.2. Study design, randomization, and blinding

This phase 3, placebo-controlled, parallel-group study consisted of 5 phases (Fig. 1): screening phase of up to 14 days; 8-week open-label, run-in (RI) phase; 6-week open-label stabilization (ST) phase; double-blind (DB) treatment phase of variable length; and a 6-month open-label extension phase. Data from the open-label extension phase will be reported separately.

During the 8-week RI phase, eligible patients received flexibly-dosed pali ER once daily in a dose range of 3 to 12 mg: an initial dose of 6 mg was gradually increased by 3 mg/day after 5 days and decreased as deemed necessary by the investigator based on patient's tolerability. Patients were recommended to be hospitalized for 8 days from the start of this phase and followed as outpatients thereafter if the investigator judged them not to be of significant risk for suicidal or violent behavior and their Clinical Global Impression — Severity (CGI-S) score was 4 (moderately ill) or less. Only those patients capable of maintaining a stable dose regimen in the last week of this phase and with PANSS score <70, and prespecified individual PANSS scores (P1 [delusions], P2 [conceptual disorganization], P3 [hallucinatory behavior], P6 [suspiciousness/persecution], P7 [hostility], and G8 [uncooperativeness]) ≤4 were eligible to enter the ST phase during which they received the established fixed dose of pali ER.

Patients who completed the RI and ST phases of the study and met the following criteria, entered the DB phase: no changes in dose in ST phase, no deliberate self-injury or violent behavior resulting in clinically significant injury to self or another person or property damage, no psychiatric hospitalization (involuntary or voluntary admission to a psychiatric hospital for decompensation of the patient's schizophrenic symptoms, PANSS score < 70) and prespecified individual PANSS scores (P1, P2, P3, P6, P7 and G8) ≤4. Patients were randomized 1:1 to receive either pali ER (at the previously established dose) or placebo via an online interactive web-based response system and/or interactive voice response system.

2.3. Study medication

In the DB phase, pali ER or matching placebo tablets were overencapsulated; pali ER tablets were provided at once daily dose of 3, 6, 9 or 12 mg. These doses were derived from 3 or 6 mg tablets with 1 to 2 tablets depending on dose. In the RI and ST phases, pali ER tablets without overencapsulation were provided.

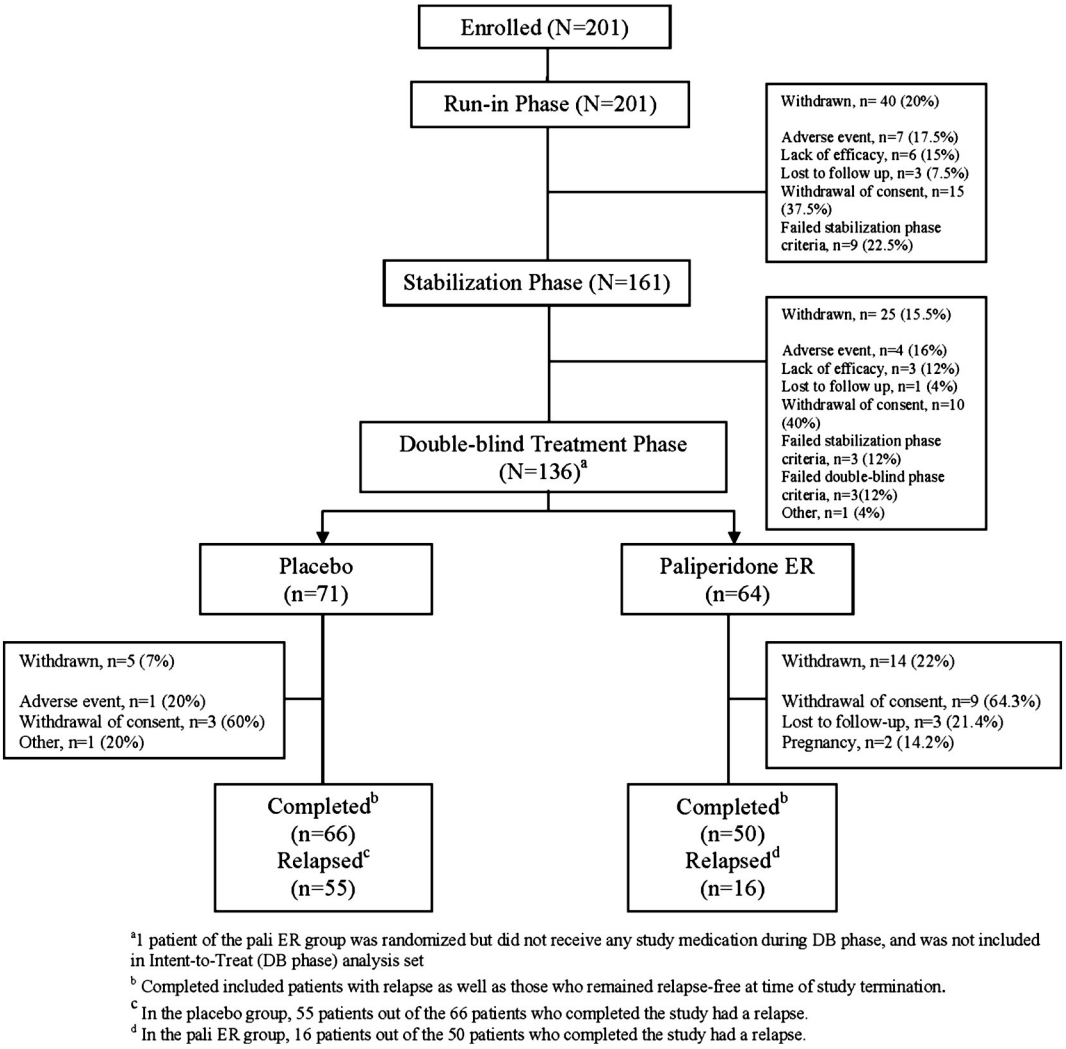
2.4. Efficacy

The primary efficacy endpoint was the time-to-first relapse during the DB phase. Relapse was defined as one or more of the following: (1) hospitalization for symptoms of schizophrenia (involuntary or voluntary admission), (2) deliberate self-injury or violent behavior, or suicidal or homicidal ideation that was clinically significant, (3) 25% increase in PANSS total score for patients who scored >40 at randomization, or a 10-point increase for patients who scored ≤40 at randomization for two consecutive assessments (within 1 week), and (4) increase in prespecified individual PANSS items scores (P1, P2, P3, P6, P7 and G8) to ≥5 for patients whose score was ≤3 at randomization, or to ≥6 for patients whose score was 4 at randomization for two consecutive assessments (within 1 week). Secondary efficacy endpoints included change from double-blind baseline to endpoint in PANSS total score, CGI-S and Personal and Social Performance Scale (PSP).

2.5. Safety

Safety assessments included extrapyramidal symptom (EPS) rating scales (Abnormal Involuntary Movement Scale [AIMS], Barnes Akathisia Rating Scale [BARS], the Simpson–Angus Rating Scale [SAS]), treatment-emergent adverse events (TEAEs), clinical laboratory tests, 12-lead electrocardiograms, vital signs measurement and physical examination findings.

The Columbia Suicide Severity Rating Scale (C-SSRS) was administered to assess for suicidal ideation and behavior.



[a] 1 patient of the pali ER group was randomized but did not receive any study medication during DB phase, and was not included in Intent-to-Treat (DB phase) analysis set

[b] Completed included patients with relapse as well as those who remained relapse-free at time of study termination.

[c] In the placebo group, 55 patients out of the 66 patients who completed the study had a relapse.

[d] In the pali ER group, 16 patients out of the 50 patients who completed the study had a relapse.

Fig. 1. Study flow and patient disposition.

2.6. Statistical analyses

2.6.1. Sample size determination

A total of 360 patients (180 per group) were planned to be enrolled in the RI phase assuming that 50% of patients would drop out from study. This was expected to provide at least 86 relapse events and to provide the study with 90% power, at the 2-sided significance level of 0.05. The median time to relapse was assumed as 11.6 months for placebo and 24.1 months for pali ER (relative risk = 0.48). The Wang and Tsiatis power boundary with shape parameter 0.15 was used for sequential monitoring (Wang and Tsiatis, 1987). An interim analysis using an Independent Data Monitoring Committee (IDMC) was pre-planned to be performed at a significance level of 0.011 when exactly 43 events were obtained. If the result of the interim analysis was significant, the DB phase had to be terminated and pali ER be declared superior to placebo in delaying time to relapse. Otherwise, the DB phase was to be continued until a total of 86 events were obtained, and a final analysis be performed at a two-sided significance level of 0.045.

2.6.2. Primary endpoint analysis

The time to relapse was determined as the time from randomization to the first relapse event in the DB phase. All patients who did not have a relapse of symptoms (who withdrew without relapse, who was symptomatically stable without relapse when study was terminated or who died before relapse) were treated as censored observations and time to censoring was determined as the time from randomization to last dose in the DB phase. Kaplan–Meier method was used to assess the primary efficacy variable (time-to-relapse), and the log-rank test (two-sided) was used to compare treatment differences.

2.6.3. Secondary endpoint analyses

All secondary efficacy analyses were performed at the 0.05 level (two-sided) across treatment groups with no adjustments for multiplicity, using the last observation carried forward (LOCF) approach. Using an analysis of covariance (ANCOVA) model with treatment and study site, and double-blind baseline PANSS total score as a covariate, the least squares means, p-values, 95% CIs were compared between the pali ER- versus placebo-treated patients.

2.6.4. Exploratory analyses

Cox proportional hazard models (including treatment and one covariate at a time) were used to estimate the hazards ratio (HR) and its 95% confidence interval (CI), and impact of various factors (age [18–25, 26 to 50, 50 to 65, >65 years], sex, BMI [normal < 25, overweight ≥ 25 to <30, obese ≥ 30 kg/m^2], prior hospitalization for psychosis at screening visit, duration of illness at screening visit [≤3 years, >3 years]) upon time-to-relapse results.

2.6.5. Populations assessed

Efficacy and safety analyses for the RI/ST phases used the all treated analysis set, which included all patients who received at least one dose of pali ER. All efficacy analyses in the DB phase used the intent-to-treat (ITT) analysis set, which included all randomized patients who received at least one dose in the DB phase. Safety analysis set for the DB phase was the same as the ITT analysis set.

3. Results

3.1. Patient disposition

Of the 201 patients enrolled in the RI phase, 161 entered the stabilization phase, of which, 136 (68%) patients were randomized to the DB phase (71, placebo; 65, pali ER) (Fig. 1). One patient in the pali ER group was randomized but did not receive any dose in DB phase, and hence was not included in the ITT analysis set. The demographic and baseline characteristics of the 135 patients in the ITT population were generally similar (Table 1). Patients were predominantly women (59%); mean age was 31.7 years (range: 18 to 61 years). A total of 153 (76%) patients received 1 or more psychotropic drugs before they entered the study. The most commonly used classes of psychotropic medications were the atypical antipsychotics (50%) and benzodiazepines (40%). Overall, 17% of patients were using anti-EPS medications before entering the RI phase. The most common (>10%) concomitant psychotropic medications used during the RI and ST phases were benzodiazepines (n = 55, 27%) which included lorazepam (n = 36, 18%) and clonazepam (n = 14, 7%). During the DB phase of the study, 7% of patients (10/135) received benzodiazepines with a higher percentage of patients in placebo than in pali ER group (11% vs. 3%); the most common was lorazepam (4%). On the other hand, 69 patients (51%) received concomitant psychotropic medications other than benzodiazepines. Trihexyphenidyl (total: n = 57 [42%]; pali ER: n = 31 [48%]; placebo: n = 26 [37%]) and propranolol (total: n = 14, 10%, pali ER: n = 7 [11%]; placebo: n = 7 [10%]) were the most commonly received concomitant psychotropic medications other than benzodiazepines, with a higher percentage of patients in the pali ER group compared with placebo group on EPS medications (48% vs 38%).

Nineteen (14%) of the 135 patients in ITT population discontinued from DB phase while 116 (86%) completed the study (i.e. relapsed or remained relapse-free up to 9 Nov 2012). More patients in the pali ER group (n = 14, 22%) withdrew from DB phase compared with the placebo group (n = 5, 7%), with withdrawal of consent (n = 9, 14%) as the most frequently reported reason for discontinuation from the study (Fig. 1).

3.2. Treatment exposure

The mean duration of exposure to study drug in the RI phase was 53.2 days and patients received a mean dose (mean of mean dose for individual patients) of 8.51 mg/day. During the RI phase, the mode dose of pali ER (the dose taken most frequently) was 9.0 mg/day by 38% of patients. During the ST phase, the mode daily doses of 6 mg, 9 mg, and 12 mg were received by 20%, 41%, and 35% of the patients, respectively. Only 4% of patients received the mode dose of 3 mg during the ST phase. The mean mode dose received in the RI phase was 9 mg (median duration of exposure: 56 days) and in the ST phase was 9.2 mg (median duration of exposure: 42 days). During the DB phase, the mode daily pali ER dose taken was 9 mg by 42%, 12 mg by 38%, and 6 mg by 19% patients. The mean (of mean dose for individual patients) dose and mean mode dose received by patients in the pali ER group was 9.5 mg/day each. Median duration of exposure in the DB phase was 36 days (placebo) and 102 days (pali ER). Mean duration of the study (including the RI/ST and DB phases but excluding the open-label extension phase) was 153 days.

3.3. Efficacy

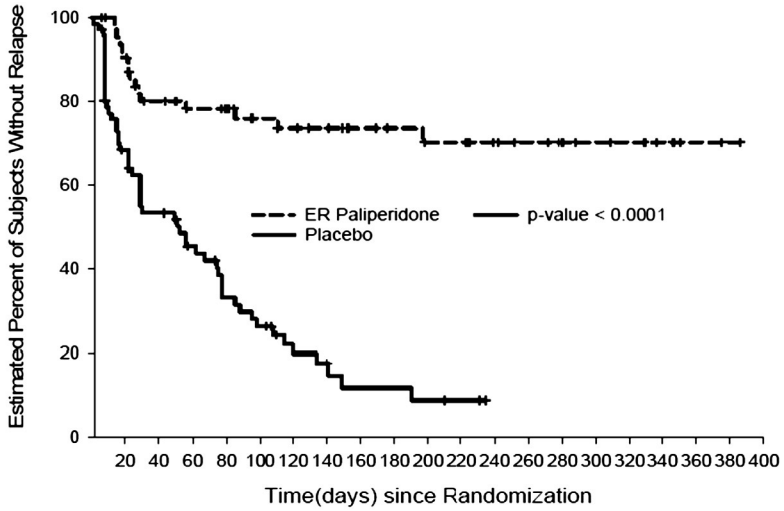
3.3.1. Primary efficacy

The interim efficacy analysis performed at the time of the 61st relapse event during the DB phase demonstrated a significant difference in time to relapse between the 2 treatment groups, in favor of pali ER compared with placebo (Chi-squared test statistic = 35.611, df = 1, $p < 0.0001$): Out of the 124 patients included in the interim ITT analysis, 75.4% (49/65) in the placebo group and 20.3% (12/59) in pali ER group experienced a relapse event. This difference exceeded the threshold for significance ($p < 0.0216$), resulting in the IDMC recommendation to stop the study early. Based on Kaplan–Meier estimates, the 25% quantile of time to relapse (estimated time point at which 25% of patients have experienced a relapse event) was 10 days in the placebo group and 111 days in the pali ER group. The median time to relapse (the estimated time point where 50% patients

Table 1
Demographic and baseline characteristics (Intent-to-Treat DB analysis set).[a]

	Placebo (n = 71)	Paliperidone ER (n = 64)	Overall (N = 135)
Sex, n (%)			
Men	30 (42)	25 (39)	55 (41)
Subpopulation, n (%)			
Chinese (Han)	69 (97)	63 (98)	132 (98)
Other	2 (3)	1 (2)	3 (2)
Age in years, mean (SD)	32.3 (12.17)	31.1 (9.36)	31.7 (10.90)
Age category, n (%)			
18–25 years	27 (38)	25 (39)	52 (39)
26–50 years	36 (51)	37 (58)	73 (54)
51–65 years	8 (11)	2 (3)	10 (7)
Run-in baseline BMI (kg/m^2), mean (SD)	23.37 (4.33)	23.32 (4.02)	23.35 (4.17)
Previous hospitalization for psychosis, n (%)	32 (45)	33 (52)	65 (48)
Duration (days) of hospitalization for psychosis prior to run-in phase, mean (SD)	81 (99.41)	89.2 (93.85)	85.2 (95.96)
DB baseline PANSS total, mean (SD)	51.5 (9.50)	53.4 (9.71)	52.4 (9.61)
DB baseline CGI-S category, n (%)			
Very Mild	22 (31)	17 (27)	39 (29)
Mild	38 (54)	27 (42)	65 (48)
Moderate	10 (14)	16 (25)	26 (19)
Marked	1 (1)	2 (3)	3 (2)
DB baseline PSP, mean (SD)	69.9 (9.37)	69.3 (11.17)	69.6 (10.23)

[a] 1 patient was randomized but did not receive any study medication during DB phase, and was not included in Intent-to-Treat (DB phase) analysis set. BMI: body mass index; SD: standard deviation; CGI-S: Clinical Global Impression — Severity; PANSS: Positive and Negative Syndrome Scale; PSP: Personal and Social Performance.



Number of Patients Left

Time	0	20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340	360	380	400
Pali ER	64	56	45	41	37	32	30	28	24	22	20	20	15	13	10	8	7	5	2	1	0
Placebo	71	46	35	27	19	15	9	6	4	4	3	2	0	0	0	0	0	0	0	0	0

[a]1 patient in the pali ER group was randomized but did not receive any study medication during DB phase, and was not included in Intent-to-Treat (DB phase) analysis set. Vertical line "|" represents time points where patients were censored

Fig. 2. Kaplan–Meier plot of time to relapse (Intent-to-Treat double-blind analysis set[a]).

experienced a relapse event) in the placebo group was 49 days, and the time was not estimable in the pali ER group because less than 50% of patients experienced a relapse event.

Final analysis results (n = 135) were confirmatory; there was a significant difference (Chi-squared test statistic = 39.443, df = 1, p < 0.0001) between the two treatment groups in the time to relapse in favor of pali ER (Fig. 2). More patients in the placebo group (n = 55, 77.5%) experienced relapse compared with those in pali ER group (n = 16, 25%) (Table 2). Based on Kaplan–Meier estimates, the 25% quantile of time to relapse was 15 days in the placebo group and 111 days in the pali ER group. The instantaneous risk (hazard) of relapse of schizophrenia symptoms was 5.23 (95% CI: 2.96, 9.25) times higher for a patient switching to placebo than for a patient continuing to receive pali ER. The most common reasons for relapse included increase in the PANSS total score value and changes (worsening) in PANSS items scores (Table 2). Most patients in both pali ER and placebo groups were clinically managed without the need for psychiatric hospitalization. Patients who experienced

Table 2
Time to relapse during the double-blind phase and frequency distribution of relapse types (Intent-to-Treat DB analysis set).[a]

	Interim analysis			Final analysis		
	Placebo (n = 65)	Paliperidone ER (n = 59)	Overall (N = 124)	Placebo (n = 71)	Paliperidone ER (n = 64)	Overall (N = 135)
Total no. of patients with relapse, n (%)	49 (75.4)	12 (20.3)	61 (49.2)	55 (77.5)	16 (25.0)	71 (52.6)
25% quantile (95% CI)	10.0 (8.0; 22.0)	111.0 (23.0; NA)	22.0 (15.0; 29.0)	15.0 (8.0; 24.0)	111.0 (26.0; NA)	22.0 (17.0; 29.0)
75% quantile (95% CI)	98.0 (77.0; 141.0)	(NA; NA)	(NA; NA)	108.0 (77.0; 141.0)	(NA; NA)	(NA; NA)
P-value[b]			<0.0001			<0.0001
Reasons for relapse, n (%)						
Psychiatric hospitalization	4 (6)	0	4 (3)	4 (6)	1 (2)	5 (4)
Psychiatric hospitalization	4 (6)	0	4 (3)	4 (6)	1 (2)	5 (4)
PANSS[c]	46 (71)	11 (19)	57 (46)	52 (73)	15 (23)	67 (50)
10 point increase in total PANSS score	3 (5)	3 (5)	6 (5)	3 (4)	3 (5)	6 (4)
Increase of 25% in total PANSS score	43 (66)	8 (14)	51 (41)	49 (69)	12 (19)	61 (45)
Deliberate self-injury, violent behavior	3 (5)	1 (2)	4 (3)	4 (6)	1 (2)	5 (4)
Deliberate self-injury	1 (2)	1 (2)	2 (2)	1 (1)	1 (2)	2 (1)
Violent behavior	3 (5)	1 (2)	4 (3)	4 (6)	1 (2)	5 (4)
Suicidal or homicidal ideation[d]	2 (3)	1 (2)	3 (2)	2 (3)	1 (2)	3 (2)
Homicidal ideation	1 (2)	0	1 (1)	1 (2)	0	1 (1)
Suicidal ideation	1 (2)	1 (2)	2 (2)	1 (2)	1 (2)	2 (2)
Suicide attempt	1 (2)	1 (2)	2 (2)	1 (2)	1 (2)	2 (2)
PANSS items (P1, P2, P3, P6, P7, G8)[a]	19 (29)	1 (2)	20 (16)	22 (31)	1 (2)	23 (17)
Score ≥5 for 2 days	16 (25)	1 (2)	17 (14)	19 (27)	1 (2)	20 (15)
Score ≥6 for 2 days	3 (5)	0	3 (2)	3 (4)	0	3 (2)

[a] 1 patient was randomized but did not receive any study medication during DB phase, and was not included in Intent-to-Treat (DB phase) analysis set.
[b] Log rank test.
[c] Regression analysis of survival data based on Cox proportional hazards model with treatment as a factor.
[d] Based on two consecutive assessments performed within one week.

relapse entered into a 6-month open-label extension phase and were treated with pali ER.

3.3.2. Secondary efficacy

Schizophrenia symptoms improved from RI baseline to endpoint (RI/ST). In the DB phase, mean (SD) total PANSS scores significantly worsened (compared with DB baseline) in placebo-treated patients (n = 71) (16.9 [16.16]) while they remained relatively stable in the pali ER-treated patients (n = 64) (2 [12.67]). Based on ANCOVA model with treatment (placebo, pali ER) and center as factors, and baseline value as a covariate, the least squares means difference of pali ER minus placebo for change in PANSS total score from DB baseline to DB end point was −14.3 (SE = 2.49) (F test = 32.83, df = [1, 115], p < 0.001 [two-sided]), which showed a significant difference between the two groups in favor of pali ER compared with placebo (Fig. 3).

During the RI and ST phases, patients' PANSS subscale and factor scores improved. In the DB phase, although the mean PANSS factor scores worsened (compared with DB baseline) in both treatment groups, the placebo group had a greater mean change (increase) at DB endpoint than the pali ER group for all factor scores, except for "disorganized thought", which showed no change in the pali ER group (Table 3).

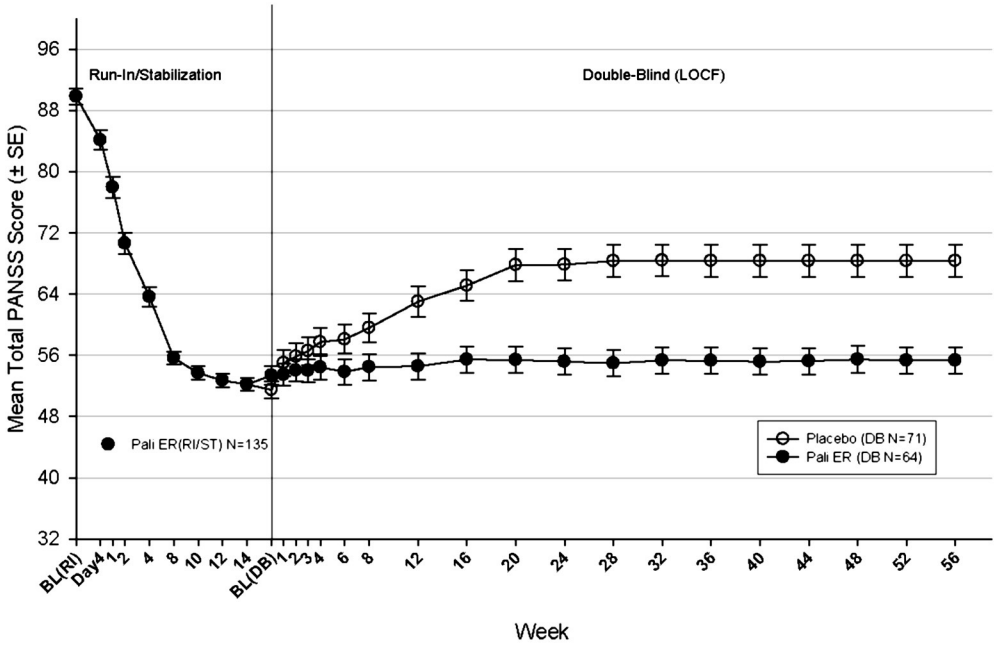
The psychotic condition of patients improved on the CGI-S scale during the RI and ST phases; only 34 (17.0%) out of 157 patients had a CGI-S score of ≥5 at the end of the ST phase, while it was 78.1% at RI baseline. At the end of DB phase, more patients in pali ER group (68.8%), as compared with the placebo group (31.0%), had severity scores of 'mild', 'very mild', or 'not ill' in CGI-S scores, which showed the bigger improvement of CGI-S scores in the pali ER group. Based on the analysis results of ranks of CGI values using ANCOVA with treatment (placebo, pali ER) and center as factors, and baseline value as a covariate, pali ER was significantly superior to placebo on the rank of changes in CGI value from DB baseline to DB end point (F test = 39.82, df = [1,115], p < 0.001 [two-sided]) (Table 3).

Mean (SD) PSP total scores improved from RI baseline (43.6 [13.96]) to RI/ST endpoint (64.3 [14.78]). The least squares mean difference of pali ER minus placebo for change in PSP total score from DB baseline to DB end point was 7.1 (SE = 2.34) (F test = 9.18, df = [1,115], p = 0.003 [two-sided]), which showed a significant difference between two groups in favor of pali ER compared with placebo (Table 3).

3.4. Safety

Overall, 81.1% of the 201 patients experienced at least 1 TEAE during the RI and ST phases; the most common TEAEs (≥10% patients) included akathisia (25.9%), tremor (11.9%), and insomnia (10.4%). During DB phase, more patients in placebo group (40.8% [n = 29/71]) experienced TEAEs compared with pali ER group (32.8% [n = 21/64]). The most common (≥10% patients) TEAEs were related to psychiatric disorders, which occurred in 21.1% placebo-treated patients and 10.9% pali ER-treated patients in pali ER group (Table 4).

A total of 12 patients (6%) experienced TEAEs that resulted in pali ER discontinuation during the RI and ST phases, of which 5 patients (2.5%) discontinued due to TEAEs related to psychiatric disorders (one patient had restlessness, irritability, synesthetic hallucination and anxiety, one patient each had restlessness, depressive symptom, psychotic disorder, and suicidal ideation). Two patients discontinued the study due to akathisia. All other TEAEs leading to study discontinuation (epilepsy, nausea, constipation, amenorrhea, QT prolongation) were reported in 1 patient each. One patient of the placebo group discontinued treatment in DB phase due to tardive dyskinesia. No deaths were reported during the RI and ST phases, although one incidence of suicidal death (in placebo group) was reported during the DB phase. Serious TEAEs were reported in 2 patients (1%) during the RI and ST phases: psychotic disorder and schizophrenia (each reported in 1 patient; 0.5%). During the DB phase, serious TEAEs were reported more frequently in placebo



BL=baseline; DB=double-blind; LOCF=last observation carried forward; PANSS=positive and negative symptom scale; RI=run-in phase; SE=standard error; ST=stabilization phase. Mean PANSS total scores are shown during the run-in and stabilization phases and the PANSS total scores over time (LOCF) during the double-blind phase.

Fig. 3. PANSS total score over time (run-in, stabilization and double-blind phases).

Table 3
Change in secondary efficacy measures from double-blind baseline to end of the double-blind phase (Intent-to-Treat DB analysis set).[a]

	Placebo (n = 71)	Paliperidone ER (n = 64)
PANSS total score, mean (SD)		
Baseline	51.5 (9.50)	53.4 (9.71)
Change from baseline	16.9 (16.16)	2 (12.67)*
PANSS subscale scores, mean (SD)		
Positive subscale		
Baseline	10.5 (3.28)	10.6 (3.13)
Change from baseline	6.1 (5.83)	1.0 (4.26)
Negative subscale		
Baseline	15.3 (4.03)	15.9 (4.43)
Change from baseline	2.3 (4.57)	0.3 (3.97)
General psychopathology		
Baseline	25.6 (4.96)	26.8 (5.27)
Change from baseline	8.5 (8.57)	0.7 (7.22)
PANSS factor scores, mean (SD)		
Positive symptoms		
Baseline	14.2 (3.77)	14.5 (3.58)
Change from baseline	5.6 (5.45)	0.5 (4.88)
Negative symptoms		
Baseline	14.7 (4.14)	15.0 (4.19)
Change from baseline	2.4 (5.15)	0.3 (4.26)
Disorganized thoughts		
Baseline	11.8 (2.91)	12.4 (3.39)
Change from baseline	3.3 (4.07)	0.0 (3.03)
Uncontrolled hostility/excitement		
Baseline	5.2 (1.93)	5.4 (2.14)
Change from baseline	3.8 (4.07)	0.6 (3.01)
Anxiety/depression		
Baseline	5.6 (2.20)	6.1 (2.51)
Change from baseline	1.8 (3.32)	0.5 (3.03)
CGI-S score, mean (SD)		
Baseline	2.9 (0.70)	3.0 (0.88)
Change from baseline	1.1 (1.11)	0.1 (0.91)#
PSP score, mean (SD)		
Baseline	69.9 (9.37)	69.3 (11.17)
Change from baseline	−10.7 (14.99)	−2.9 (12.87)**

SD: standard deviation; CGI-S: Clinical Global Impression — Severity; PANSS: Positive and Negative Syndrome Scale; PSP: Personal and Social Performance.
$p < 0.001$.
* $p < 0.0001$.
** $p < 0.003$.
[a] 1 patient was randomized but did not receive any study medication during DB phase, and was not included in Intent-to-Treat (DB phase) analysis set.

Table 4
Treatment-emergent adverse events in at least 2% of patients in either treatment group (Intent-to-Treat DB analysis set).[a]

	Placebo (n = 71) n (%)	Pali ER (n = 64) n (%)
Total number of patients with TEAEs	29 (41)	21 (33)
Schizophrenia	6 (9)	2 (3)
Insomnia	6 (9)	1 (2)
Aggression	2 (3)	3 (5)
Akathisia	0	3 (5)
Nausea	1 (1)	2 (3)
Tremor	0	2 (3)
Salivary hypersecretion	0	2 (3)
Amenorrhea	0	2 (3)
Agitation	2 (3)	0
Weight decreased	2 (3)	0
Nasopharyngitis	2 (3)	0
Oropharyngeal pain	2 (3)	0

[a] 1 patient was randomized but did not receive any study medication during DB phase, and was not included in Intent-to-Treat (DB phase) analysis set. TEAEs: treatment-emergent adverse events.

(11%) than pali ER-treated patients (3%); all serious TEAEs were psychiatric-related.

The most common EPS-related TEAEs reported in the RI and ST phases were those grouped under hyperkinesia (31.3%), parkinsonism (12.4%), and tremor (11.9%). In the DB phase, the incidence of EPS-related TEAEs was higher in the pali ER group (9.4%; hyperkinesia [6.3%] and tremor [3.1%]) compared with the placebo group (4.2%; dyskinesia [2.8%] and parkinsonism [1.4%]): none of them were serious. This difference in EPS-related TEAEs is corroborated by the higher proportion of patients in the pali ER group (48%) taking anti-EPS medications compared with placebo (38%) in the DB phase.

Four patients experienced prolactin-related TEAEs in the RI and ST phase. Two patients experienced loss of libido, 1 experienced galactorrhea and 1 sexual dysfunction. Three women experienced irregular menstruation and 2 women had amenorrhea; in DB phase, 1 woman (pali ER group) experienced galactorrhea, 2 women (pali ER group) experienced amenorrhea and 1 woman of each group experienced irregular menstruation. During the RI and ST phases, prolactin levels increased from baseline to the end of the RI phase in both men and women; thereafter, they remained elevated at the same level throughout the ST phase (mean [SD] change from RI baseline to RI/ST endpoint: men: 25.6 [26.06] μg/L; women: 78.8 [73.39] μg/L). Mean prolactin levels of both men and women in the placebo group decreased from DB baseline levels to levels within the reference range at the end of the DB phase (change in mean [SD], men: −33.98 [15.99] μg/L; women: −90.33 [53.58] μg/L). On the other hand, in the pali ER group, there was a gradual increase in mean prolactin levels from the DB baseline during the first 12 weeks of the DB phase (change in mean [SD], men: 2.95 [8.31] μg/L; women: −12.98 [73.76] μg/L) followed by a gradual decrease or stabilization for the remaining course of the study (change in mean [SD], men: −5.59 [14.35] μg/L; women: −24.40 [74.93] μg/L). Two patients experienced glucose-related TEAEs during RI and ST phases while none did during the DB phase in either treatment group.

TEAEs related to suicidality were reported for 1 patient (suicidal ideation) during the RI and ST phases, and for 2 patients (suicide attempt [pali ER group] and completed suicide [placebo], one each) during the DB phase. A majority of patients (98%) had a maximum C-SSRS score of 0 (no suicidal ideation), and 2% of patients had a score of 1 (wish to be dead).

There were markedly abnormal increases in pulse rate, both standing (14.2% patients) and supine (6.5% patients), with markedly abnormal decreases in standing (8.2% patients) and supine systolic blood pressure measurements (5.5% patients) in the RI and ST phases. During the DB phase, more patients in the pali ER group had increased standing pulse rate (pali ER 10% vs placebo 3%) and decreased supine systolic blood pressure (pali ER 5% vs. placebo 0%). No cases of QTc values exceeding 500 ms were reported at any time during the study. Of the 200 patients with normal QTcLD values at baseline, 2 had QTcLD values of >450 ms and 1 had QTcLD values of >480 ms during the RI and ST phases. In the DB phase, QTcLD values shifted from normal to >450 ms in 1/56 patients in the pali ER group only. There were no instances of shifts from normal to >480 ms or >500 ms QTcLD interval in either treatment group. The QTcLD values worsened more in the placebo group than in the pali ER group with mean (SD) change in QTcLD values from DB baseline to DB endpoint to be −6.4 (15.05) ms in the placebo group and −1.8 (16.70) ms in the pali ER group. Body weight increase was reported as an AE in the open-label RI and ST phases (6.5%) and DB phase (placebo 1.4% vs. pali ER 1.6%). In the DB phase, body weight increases were more prevalent and greater in pali ER group (42%, mean [SD] increase: 3.90 [5.77] kg) versus placebo (30%, mean [SD] increase: 2.05 [4.85] kg).

4. Discussion

The current study demonstrated that flexibly dosed pali ER treatment (3–12 mg/day) significantly delayed relapse in Chinese

patients with schizophrenia. The study was terminated early on recommendations from the IDMC because the primary efficacy endpoint (time to relapse) was met at the preplanned interim analysis. The final statistical analysis findings corroborated those of the interim analysis. Additionally, results from an exploratory analysis further suggested that the efficacy of pali ER with regard to time to relapse of symptoms of schizophrenia was consistent regardless of BMI, sex, age, prior hospitalization status, or duration of illness ($p < 0.0001$ for all factors). Treatment with pali ER was associated with an overall improvement in symptoms of schizophrenia as observed with significant reductions in PANSS total scores, individual PANSS factor scores, CGI-SCH scores and increment in PSP scores at the end of DB phase.

The study showed that 25% of pali ER-treated patients had relapse; these results are consistent with a global long-term pali ER study with a similar design wherein 22% of pali ER-treated patients had relapse at final analysis (Kramer et al., 2007). Our results are also consistent with a recent meta-analysis of data of 6493 patients with schizophrenia from 65 clinical trials, which demonstrated a relapse in 25% of patients treated with antipsychotic drugs compared with 69% of patients treated with placebo (Leucht et al., 2012). This suggests pali ER's efficacy in preventing relapse is similar in this Chinese population as within a more diversified population.

Interestingly, relapse rates for placebo were higher in the current study (77.5%) compared with the global long-term pali ER study (53%) (Kramer et al., 2007). The difference in relapse rates in the placebo-treated group in our study may be due higher dose received during the run-in/stabilization phase in the global long-term pali ER study (Kramer et al., 2007) compared with this study. Other possible reasons, such as differing patient expectations about clinical trial treatments or even standards of care between countries may also have contributed to the difference.

Patients who continued on pali ER treatment from RI and ST phases to DB phase had relatively stable PANSS scores over the 56 week period, compared with those who switched to placebo, showing that symptom stability was maintained. Improvements were also observed for all of the five symptomatic domains (PANSS Marder factor scores) of schizophrenia: positive symptoms, negative symptoms, depression/anxiety, uncontrolled hostility/excitement and disorganized thoughts. Robust symptom improvement in pali-ER-treated patients was further reflected in the improvements in CGI-S scores. Categorical analysis of patients at baseline and DB end point demonstrated improvement in the PSP scores by at least 1 category in pali ER-treated patients compared with placebo. The results of secondary efficacy evaluations were thus aligned with the primary findings, supporting pali ER's efficacy in this population, and were also consistent with results from the global long term pali ER study with similar design (Kramer et al., 2007). Overall, these findings also corroborate data from other short- and long-term studies for pali ER (Canuso et al., 2009; Davidson et al., 2007; Emsley et al., 2008a; Kane et al., 2007; Kim et al., 2013; Kramer et al., 2007; Schmauss et al., 2012).

Consistent with the global long-term pali ER studies, the results of this study also suggest that continued benefit is realized with extended treatment. Substantial improvement occurred in symptom severity during the 14-week open-label RI and ST phases, and symptom stability was achieved. Subsequently, within the first 2 weeks of the DB phase, there was a statistically significant separation in the number of efficacy and quality-of-life measures between placebo- and pali ER-treated patients. Pali ER-treated patients remained relatively stable, whereas notable worsening of schizophrenia symptoms occurred in placebo-treated patients, thus indicating that pali ER treatment maintained the significant symptom control achieved during the ST period.

The beneficial effects following pali ER treatment may be a result of supersensitivity psychosis; however, it is now shown that abrupt or gradual withdrawal of prior treatment with antipsychotic drugs in the placebo group does not change the relapse risk (Leucht et al., 2012).

Safety findings in this population were consistent with the earlier pali ER studies (Davidson et al., 2007; Emsley et al., 2008b; Kane et al., 2007; Kramer et al., 2007, 2010). The incidence of TEAEs and discontinuation due to TEAEs was low in pali ER-treated patients compared with placebo-treated patients. One case of suicidal ideation during the RI/ST phase and 2 cases of suicide attempt during DB phase were reported for the pali ER group.

Consistent with the known pharmacology of pali ER, there were increases in plasma prolactin levels, which were more pronounced in women than men. Similar to the earlier pali ER short- and long-term studies, few prolactin-related TEAEs were reported and no worsening of any other metabolic parameter was observed in this study (Kane et al., 2007; Kramer et al., 2007).

The incidence of EPS-related TEAEs observed in this study is also comparable to the global pali ER study (Kramer et al., 2007). None of the EPS-related TEAEs in this trial were considered serious. EPS-related TEAEs occurred more often in pali ER- than placebo-treated patients, and more pali ER-treated patients required anti-EPS medications in the DB phase. This may also be attributed to the fact that investigators usually tend to titrate upward from the starting dose of 6 mg/day and rarely reduce the dose even when efficacy is achieved in absence of any adverse event.

Glucose-related TEAEs were uncommon (1%) during the RI and ST; there were no reports of glucose-related TEAEs during the DB phase consistent with the absence of clinically meaningful changes from baseline in mean glucose levels. The incidence of TEAEs of weight increased, which represents patients with clinically significant weight gain as assessed by the investigators, was low in both the RI and ST phases (6.5%), and DB phase (placebo 1.4% vs. pali ER 1.6%). The incidence of treatment-emergent abnormalities in recorded ECG parameters (PR and QRS intervals) was also low during the course of the study and showed no clinically relevant differences between the two treatments during the DB phase.

The study thus extended the findings from the earlier pali ER relapse-prevention study and demonstrated that pali ER treatment (3–12 mg) was similarly efficacious in preventing relapse and generally tolerable in Chinese patients with schizophrenia.

Author contributions

Drs. Wang, Nuamah, Wu, Y., Wu, Q., Rui and Gopal and Ms. Liu were involved in design and analysis. Drs. Wu, Y. and Wu, Q. were involved in data collection. Dr. Shu was the principal investigator for the study.

Disclosures

Drs. Wu, Y., Wu, Q., Nuamah, Rui and Gopal, and Ms. Liu are employees of Janssen Research & Development. Dr. Wang was an employee of Janssen Research & Development, China during the conduct and analyses of this study. Dr. Shu was the principal investigator for the study and is employed by the Peking University Sixth Hospital, Beijing, China.

All authors meet ICMJE criteria and all those who fulfilled those criteria are listed as authors. All authors had access to the study data and made the final decision about where to publish these data.

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References

An FR, Xiang YT, Wang CY, Zhang GP, Chiu HF, Chan SS, et al. Change of psychotropic drug prescription for schizophrenia in a psychiatric institution in Beijing, China between 1999 and 2008. Int J Clin Pharmacol Ther 2010;48(4):270–4.

Banerjee A. Cross-cultural variance of schizophrenia in symptoms, diagnosis and treatment. Georgetown Undergrad J Health Sci 2012;6:18–24.

Bhugra D, Bhui K. Ethnic and cultural factors in psychopharmacology. Adv Psychiatr Treat 1999;5:89–95.

Canuso CM, Bossie CA, Turkoz I, Alphs L. Paliperidone extended-release for schizophrenia: effects on symptoms and functioning in acutely ill patients with negative symptoms. Schizophr Res 2009;113(1):56–64.

Coppola D, Liu Y, Gopal S, Remmerie B, Samtani MN, Hough DW, et al. A one-year prospective study of the safety, tolerability and pharmacokinetics of the highest available dose of paliperidone palmitate in patients with schizophrenia. BMC Psychiatry 2012;12:26.

Csernansky JG, Mahmoud R, Brenner R, Risperidone USASG. A comparison of risperidone and haloperidol for the prevention of relapse in patients with schizophrenia. N Engl J Med 2002;346(1):16–22.

Davidson M, Emsley R, Kramer M, Ford L, Pan G, Lim P, et al. Efficacy, safety and early response of paliperidone extended-release tablets (paliperidone ER): results of a 6-week, randomized, placebo-controlled study. Schizophr Res 2007;93(1–3):117–30.

Emsley R, Berwaerts J, Eerdekens M, Kramer M, Lane R, Lim P, et al. Efficacy and safety of oral paliperidone extended-release tablets in the treatment of acute schizophrenia: pooled data from three 52-week open-label studies. Int Clin Psychopharmacol 2008a;23(6):343–56.

Emsley R, Oosthuizen P, Koen L, Niehaus DJ, Medori R, Rabinowitz J. Remission in patients with first-episode schizophrenia receiving assured antipsychotic medication: a study with risperidone long-acting injection. Int Clin Psychopharmacol 2008b;23(6):325–31.

Kane JM, Carson WH, Saha AR, McQuade RD, Ingenito GG, Zimbroff DL, et al. Efficacy and safety of aripiprazole and haloperidol versus placebo in patients with schizophrenia and schizoaffective disorder. J Clin Psychiatry 2002;63(9):763–71.

Kane J, Canas F, Kramer M, Ford L, Gassmann-Mayer C, Lim P, et al. Treatment of schizophrenia with paliperidone extended-release tablets: a 6-week placebo-controlled trial. Schizophr Res 2007;90(1–3):147–61.

Kim EY, Chang SM, Shim JC, Joo EJ, Kim JJ, Kim YS, et al. Long-term effectiveness of flexibly dosed paliperidone extended-release: comparison among patients with schizophrenia switching from risperidone and other antipsychotic agents. Curr Med Res Opin 2013;29(10):1231–40.

Kramer M, Simpson G, Maciulis V, Kushner S, Vijapurkar U, Lim P, et al. Paliperidone extended-release tablets for prevention of symptom recurrence in patients with schizophrenia: a randomized, double-blind, placebo-controlled study. J Clin Psychopharmacol 2007;27(1):6–14.

Kramer M, Simpson G, Maciulis V, Kushner S, Liu Y, Lim P, et al. One-year open-label safety and efficacy study of paliperidone extended-release tablets in patients with schizophrenia. CNS Spectr 2010;15(8):506–14.

Lehman AF, Lieberman JA, Dixon LB, McGlashan TH, Miller AL, Perkins DO, et al. Practice guideline for the treatment of patients with schizophrenia, second edition. Am J Psychiatry 2004;161(2 Suppl.):1–56.

Leucht S, Tardy M, Komossa K, Heres S, Kissling W, Salanti G, et al. Antipsychotic drugs versus placebo for relapse prevention in schizophrenia: a systematic review and meta-analysis. Lancet 2012;379(9831):2063–71.

Li W, Song X, Zhang H, Yang Y, Jiang C, Xiao B, Li W, Yang G, Zhao J, Guo W, Lv L. Association study of RELN polymorphisms with schizophrenia in Han Chinese population. Prog Neuropsychopharmacol Biol Psychiatry 2011;35(6):1505–11.

Schmauss M, Jukic V, Siracusano A, Bidzan L, Badescu GM, Maciulis V, et al. Flexible dosing with paliperidone ER in the treatment of patients with acutely exacerbated schizophrenia: results from a single-arm, open-label study. Curr Med Res Opin 2012;28(8):1395–404.

Schooler N, Rabinowitz J, Davidson M, Emsley R, Harvey PD, Kopala L, et al. Risperidone and haloperidol in first-episode psychosis: a long-term randomized trial. Am J Psychiatry 2005;162(5):947–53.

Taylor DM. Aripiprazole: a review of its pharmacology and clinical use. Int J Clin Pract 2003;57(1):49–54.

Versola-Russo JM. Cultural and demographic factors of schizophrenia. Int J Psychosoc Rehabil 2006;10:89–103.

Wang SK, Tsiatis AA. Approximately optimal one-parameter boundaries for group sequential trials. Biometrics 1987;43(1):193–9.

Williams DR, Earl TR. Commentary: race and mental health — more questions than answers. Int J Epidemiol 2007;36(4):758–60.